

Jieke Shi

Research Scientist | NRF Postdoctoral Award Recipient
Singapore Management University

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RISE Center, 10 Canning Rise, Singapore 179873

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RESEARCH INTERESTS

Software Engineering, Program Synthesis, Software Testing, Code LLMs

I design and implement **automated synthesis**-assisted methods for validating the **safety** of software systems and enhancing the **efficiency** of large language models serving.

EDUCATION

Aug 2022 – Mar 2025 **Singapore Management University**, School of Computing and Information Systems
Ph.D. Candidate in Computer Science

- **38 Conference/Journal Publications**
- **360+ Bugs/Vulnerabilities Detected and Fixed** in widely-used open-source projects (e.g., rustc, GCC, OpenSSL, curl, FFmpeg, OP-TEE)
- **National Research Foundation (NRF) Postdoctoral Award** (\$250,000)
- **ACM SIGSOFT Distinguished Paper Awards** (ASE'22 nomination, ASE'25 winner, ICSE'26 winner)
- **IEEE Computer Society TCSE Distinguished Paper Award** (SANER 2026)
- **IEEE Computer Society Best Paper Award** (TSE 2025)
- **SMU Research Staff Excellence Awards** (2025)
- **SMU SCIS Dean's List Award** (2023 & 2025)
- **SMU Presidential Doctoral Fellowship** (2023)
- **Honorable Mention Award** (ACSAC 2022)

Supervisor: ACM/IEEE/ASE Fellow, OUB Chair Professor [David Lo](#)

Aug 2016 – Jun 2020 **Yangzhou University**, School of Information Engineering
B.Eng. in Electronic Engineering

- **Outstanding Undergraduate Thesis Award** (Top 0.2% in province)
- **Outstanding Graduate** (Top 5%)

EMPLOYMENT

Mar 2026 – Present **Singapore Management University**, Research Scientist
Awarded the National Research Foundation (NRF) Postdoctoral Award, which provides up to 4 years of funding with a SGD 250,000 research grant to support independent research.

Jun 2021 – Mar 2026 **Singapore Management University**, Research Engineer
Working as lead research staff under the AI.SG Programme on the testing of Reinforcement Learning (RL)-enabled systems. Led to multiple publications covering runtime shielding and RL system safety validation, including [J.6, J.7, C.5, C.9, C.10].

Jul 2020 – Apr 2021 **Chinese Academy of Sciences, Institute of Computing**, Research Assistant Intern
Developed a prototype system for generating faithful natural language text from structured data.

PUBLICATIONS

Note: In software engineering, ICSE and ASE are recognized as top-tier conferences, and TOSEM and TSE are top-tier journals (CORE A*); MSR, ICSME, and SANER are prominent conferences covering software mining, analysis, and evolution (CORE A). In security, S&P is a top-tier conference (CORE A*), while ACSAC is a leading venue with a focus on system security applications (CORE A).

[C]: Full Conference Paper (Total: 24) [J]: Full Journal Paper (Total: 8) [S]: Short Paper (Total: 6)

2026

- [C.23] Weifeng Sun, **Jieke Shi**, Zhou Yang, Yuchen Chen, Hongyan Li, Meng Yan, and David Lo. “*How Reasoning Shapes Social Bias in LLM-Generated Code?*”. In Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering (**ASE 2026**, Research Track).
- [C.22] Xiuwei Shang, Li Hu, Xiao Jiang, **Jieke Shi** (✉ **Corr.**), Junda He, Zhou Yang, Shaoyin Cheng, Guoqiang Chen, Weiming Zhang, and David Lo. “*Beyond Text Matching: Towards Reference-Free Evaluation for Human-Oriented Binary Reverse Engineering*”. In Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering (**ASE 2026**, Research Track).
- [C.21] Thong Hoang, Mykhailo Klymenko, Xiwei Xu, Shidong Pan, Yi Ding, Xushuo Tang, Zhengyi Yang, **Jieke Shi**, and David Lo. “*ESG Reporting Lifecycle Management with Large Language Models and AI Agents*”. In Proceedings of the 23rd IEEE International Conference on Software Architecture (**ICSA 2026**, Software Architecture in Practice). [[PDF](#)]
- [C.20] **Jieke Shi**, Junda He, Zhou Yang, Chengran Yang, Mykhailo Klymenko, Thong Hoang, Sherry Xu, Zhenchang Xing, and David Lo. “*Compiling Code LLMs into Lightweight Executables*”. In Proceedings of the 34th ACM International Conference on the Foundations of Software Engineering (**FSE 2026**, Research Track), 24 pages. [[PDF](#)] [[Code](#)]
- [C.19] Chengyan Ma, **Jieke Shi** (✉ **Corr.**), Ruidong Han, Ye Liu, Feng Li, Yuqing Niu, and David Lo. “*Automated Repair of TEE Partitioning Issues via DSL-Guided and LLM-Assisted Patching*”. In Proceedings of the 34th ACM International Conference on the Foundations of Software Engineering (**FSE 2026**, Research Track), 24 pages.
- [C.18] Yuqing Niu, **Jieke Shi** (✉ **Corr.**), Ruidong Han, Ye Liu, Chengyan Ma, Yunbo Lyu, and David Lo. “*What You Trust Is Insecure: Demystifying How Developers (Mis)Use Trusted Execution Environments in Practice*”. In Proceedings of the 33rd IEEE International Conference on Software Analysis, Evolution and Reengineering (**SANER 2026**, 🏆 **IEEE Computer Society TCSE Distinguished Paper Award**). [[PDF](#)] [[Code](#)]
- [C.17] Bo Wang, Yueyang Chen, **Jieke Shi**, Minghui Li, Yunbo Lyu, Yinan Wu, Youfang Lin, and Zhou Yang. “*Hidden Licensing Risks in the LLMware Ecosystem*”. In Proceedings of the 35th ACM SIGSOFT International Symposium on Software Testing and Analysis (**ISSTA 2026**, Research Track). [[PDF](#)] [[Code](#)]
- [C.24] Junda He, **Jieke Shi**, Zhou Yang, Mingfei Cheng, and David Lo. “*Learning from the Test: Self-Referential Differential Testing for Deep RL Agents*”. In Proceedings of the 35th ACM SIGSOFT International Symposium on Software Testing and Analysis (**ISSTA 2026**, Research Track).
- [S.6] Chengyan Ma, **Jieke Shi**, Ruidong Han, Ye Liu, Yuqing Niu, and David Lo. “*Finding Missing Input Validation in TEEs via LLM-Assisted Symbolic Execution*”. In Proceedings of the 3rd ACM International Conference on AI Foundation Models and Software Engineering (**FORGE 2026**, New Idea Papers).
- [C.16] Chengran Yang, Zhensu Sun, Hong Jin Kang, **Jieke Shi**, and David Lo. “*Think Like Human Developers: Harnessing Community Knowledge for Structured Code Reasoning*”. In Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (**ICSE 2026**, 🏆 **ACM SIGSOFT Distinguished Paper Award**). [[PDF](#)]

- [C.15] Huihui Huang, Ratnadira Widyasari, Ting Zhang, Ivana Clairine IRSAN, **Jieke Shi** ( **Corr.**), Han Wei Ang, Frank Liauw, Eng Lieh Ouh, Lwin Khin Shar, Hong Jin Kang, and David Lo. “Back to the Basics: Rethinking Issue-Commit Linking with LLM-Assisted Retrieval”. In Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (**ICSE 2026**). [[PDF](#)] [[Code](#)]
- [J.8] Junda He, **Jieke Shi**, Terry Yue Zhuo, Christoph Treude, Jiamou Sun, Zhenchang Xing, Xiaoning Du, and David Lo. “LLM-as-a-Judge for Software Engineering: Literature Review, Vision, and the Road Ahead”. ACM Transactions on Software Engineering and Methodology (**TOSEM 2026**). [[PDF](#)]
- [S.5] Huihui Huang, **Jieke Shi**, Junkai Chen, Ting Zhang, Yikun Li, Chengran Yang, Eng Lieh Ouh, Lwin Khin Shar, and David Lo. “PenForge: On-the-Fly Expert Agent Construction for Automated Penetration Testing”. In Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (**ICSE 2026**, NIER Track). [[PDF](#)] [[Code](#)]
- [C.14] Junkai Chen, Huihui Huang, Yunbo Lyu, Junwen An, **Jieke Shi**, Chengran Yang, Ting Zhang, Haoye Tian, Yikun Li, Zhenhao Li, Xin Zhou, Xing Hu, and David Lo. “SecureVibeBench: Evaluating Secure Coding Capabilities of Code Agents with Realistic Vulnerability Scenarios”. In Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026**, Main Conference). [[PDF](#)]
- [C.13] Chengran Yang, Ting Zhang, Jinfeng Jiang, Xin Zhou, Haoye Tian, Mingzhe Du, **Jieke Shi**, Junkai Chen, Yikun Li, Eng Lieh Ouh, Lwin Khin Shar, and David Lo. “SeCuRepair: Semantics-Aligned, Curriculum-Driven, and Reasoning-Enhanced Vulnerability Repair Framework”. In Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026**, Main Conference, Oral, 🏆 **Outstanding Paper Candidate**). [[PDF](#)]

2025

- [J.7] **Jieke Shi**, Junda He, Zhou Yang, Đorđe Žikelić, and David Lo. “Synthesizing Efficient and Permissive Programmatic Runtime Shields for Neural Policies”. ACM Transactions on Software Engineering and Methodology (**TOSEM 2025**). [[PDF](#)] [[Code](#)]
- [J.6] **Jieke Shi**, Zhou Yang, Junda He, Bowen Xu, Dongsun Kim, DongGyun Han, and David Lo. “Finding Safety Violations of AI-Enabled Control Systems through the Lens of Synthesized Proxy Programs”. ACM Transactions on Software Engineering and Methodology (**TOSEM 2025**). [[PDF](#)] [[Code](#)]
- [J.5] **Jieke Shi**, Zhou Yang, and David Lo. “Efficient and Green Large Language Models for Software Engineering: Literature Review, Vision, and the Road Ahead”. ACM Transactions on Software Engineering and Methodology (**TOSEM 2025**). [[PDF](#)]
- [J.4] Zhou Yang, **Jieke Shi** ( **Corr.**), Premkumar Devanbu, and David Lo. “Ecosystem of Large Language Models for Code”. ACM Transactions on Software Engineering and Methodology (**TOSEM 2025**). [[PDF](#)] [[Code](#)]
- [C.12] Yunbo Lyu, Zhou Yang, **Jieke Shi**, Jianming Chang, Yue Liu, and David Lo. “My productivity is boosted, but ...” Demystifying Users’ Perception on AI Coding Assistants”. In Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (**ASE 2025**, 🏆 **ACM SIGSOFT Distinguished Paper Award**). [[PDF](#)] [[Code](#)]
- [J.3] Zhou Yang, **Jieke Shi**, Muhammad Hilmi Asyrofi, Bowen Xu, Xin Zhou, DongGyun Han, and David Lo. “Prioritizing Speech Test Cases”. ACM Transactions on Software Engineering and Methodology (**TOSEM 2025**). [[PDF](#)] [[Code](#)]

2024

- [J.2] Zhou Yang, Zhipeng Zhao, Chenyu Wang, **Jieke Shi**, Dongsun Kim, DongGyun Han, and David Lo. “Gotcha! This Model Uses My Code! Evaluating Membership Leakage Risks in Code Models”. IEEE Transactions on Software Engineering (**TSE 2024**). [[PDF](#)]

- [C.11] **Jieke Shi**, Zhou Yang, Hong Jin Kang, Bowen Xu, Junda He, and David Lo. “Greening Large Language Models of Code”. In Proceedings of the 46th IEEE/ACM International Conference on Software Engineering (**ICSE 2024**). [\[PDF\]](#) [\[Code\]](#)
- [J.1] Zhou Yang, Bowen Xu, Jie M. Zhang, Hong Jin Kang, **Jieke Shi**, Junda He, and David Lo. “Stealthy Backdoor Attack for Code Models”. IEEE Transactions on Software Engineering (**TSE 2024**, 🏆 **IEEE Computer Society Best Paper Award**). [\[PDF\]](#) [\[Code\]](#)
- [C.10] Chen Gong, Zhou Yang, Yunpeng Bai, Junda He, **Jieke Shi**, Kecen Li, Arunesh Sinha, Bowen Xu, Xinwen Hou, David Lo, and Tianhao Wang. “BAFFLE: Hiding Backdoors in Offline Reinforcement Learning Datasets”. In Proceedings of the 45th IEEE Symposium on Security and Privacy (**S&P 2024**). [\[PDF\]](#) [\[Code\]](#)
- [C.9] Junda He, Zhou Yang, **Jieke Shi**, Chengran Yang, Kisub Kim, Bowen Xu, Xin Zhou, and David Lo. “Curiosity-Driven Testing for Sequential Decision-Making Process”. In Proceedings of the 46th IEEE/ACM International Conference on Software Engineering (**ICSE 2024**). [\[PDF\]](#) [\[Code\]](#)
- [C.8] Zhou Yang, Zhipeng Zhao, Chenyu Wang, **Jieke Shi**, Dongsun Kim, DongGyun Han, and David Lo. “Unveiling Memorization in Code Models”. In Proceedings of the 46th IEEE/ACM International Conference on Software Engineering (**ICSE 2024**). [\[PDF\]](#) [\[Code\]](#)

2023

- [C.7] Zhou Yang, Chenyu Wang, **Jieke Shi**, Thong Hoang, Pavneet Kochhar, Qinghua Lu, Zhenchang Xing, and David Lo. “What Do Users Ask in Open-Source AI Repositories? An Empirical Study of GitHub Issues”. In Proceedings of the 20th IEEE/ACM International Conference on Mining Software Repositories (**MSR 2023**). [\[PDF\]](#) [\[Code\]](#)

2022

- [C.6] **Jieke Shi**, Zhou Yang, Bowen Xu, Hong Jin Kang, and David Lo. “Compressing Pre-trained Models of Code into 3 MB”. In Proceedings of the 37th IEEE/ACM International Conference on Automated Software Engineering (**ASE 2022**, 🏆 **ACM SIGSOFT Distinguished Paper Award Nomination**). [\[PDF\]](#) [\[Poster\]](#) [\[Code\]](#)
- [S.4] **Jieke Shi**, Zhou Yang, Junda He, Bowen Xu, and David Lo. “Can Identifier Splitting Improve Open-Vocabulary Language Model of Code?”. In Proceedings of the 29th IEEE International Conference on Software Analysis, Evolution and Reengineering (**SANER 2022**, ERA Track). [\[PDF\]](#) [\[Poster\]](#) [\[Code\]](#)
- [C.5] Chen Gong, Zhou Yang, Yunpeng Bai, **Jieke Shi**, Arunesh Sinha, Bowen Xu, David Lo, Xinwen Hou, Guoliang Fan. “Curiosity-Driven and Victim-Aware Adversarial Policies”. In Proceedings of the 39th Annual Computer Security Applications Conference (**ACSAC 2022**, 🏆 **Honorable Mention Award**). [\[PDF\]](#) [\[Code\]](#)
- [C.4] Zhou Yang, **Jieke Shi**, Muhammad Hilmi Asyrofi, and David Lo. “Revisiting Neuron Coverage Metrics and Quality of Deep Neural Networks”. In Proceedings of the 29th IEEE International Conference on Software Analysis, Evolution and Reengineering (**SANER 2022**). [\[PDF\]](#) [\[Code\]](#)
- [C.3] Zhou Yang, **Jieke Shi**, Junda He, and David Lo. “Natural Attack for Pre-trained Models of Code”. In Proceedings of the 44th IEEE/ACM International Conference on Software Engineering (**ICSE 2022**). [\[PDF\]](#) [\[Code\]](#)
- [C.2] Chengran Yang, Bowen Xu, Ferdian Thung, Yucen Shi, Ting Zhang, Zhou Yang, Xin Zhou, **Jieke Shi**, Junda He, DongGyun Han, and David Lo. “Answer Summarization for Technical Queries: Benchmark and New Approach”. In Proceedings of the 37th IEEE/ACM International Conference on Automated Software Engineering (**ASE 2022**). [\[PDF\]](#) [\[Code\]](#)
- [C.1] Ratnadira Widayarsi, Stefanus Agus Haryono, Ferdian Thung, **Jieke Shi**, Constance Tan, Fiona Wee, Jack Phan, and David Lo. “On the Influence of Biases in Bug Localization: Evaluation and Benchmark”. In Proceedings of the 29th IEEE International Conference on Software Analysis, Evolution and Reengineering (**SANER 2022**). [\[PDF\]](#) [\[Code\]](#) [\[Dataset\]](#)

2021

- [S.3] Zhou Yang , **Jieke Shi** , Shaowei Wang, and David Lo. “*IncBL: Incremental Bug Localization*”. In Proceedings of the 36th IEEE/ACM International Conference on Automated Software Engineering (ASE 2021, Tool Demonstrations). [[PDF](#)] [[Poster](#)] [[Code](#)]
- [S.2] Zhou Yang, Harshit Jain, **Jieke Shi**, Muhammad Hilmi Asyrofi, and David Lo. “*BiasHeal: On-the-Fly Black-Box Healing of Bias in Sentiment Analysis Systems*”. In Proceedings of the 37th IEEE International Conference on Software Maintenance and Evolution (ICSME 2021, NIER Track). [[PDF](#)] [[Code](#)]
- [S.1] Muhammad Hilmi Asyrofi, Zhou Yang, **Jieke Shi**, Chu Wei Quan, and David Lo. “*Can Differential Testing Improve Automatic Speech Recognition Systems?*”. In Proceedings of the 37th IEEE International Conference on Software Maintenance and Evolution (ICSME 2021, NIER Track). [[PDF](#)] [[Code](#)]

HONORS & AWARDS

- 2026 **National Research Foundation (NRF) Postdoctoral Award** (S\$250,000), Singapore NRF
ACM SIGSOFT Distinguished Paper Award, ICSE 2026
IEEE Computer Society TCSE Distinguished Paper Awards, SANER 2026
- 2025 **ACM SIGSOFT Distinguished Paper Award**, ASE 2025
IEEE Computer Society Best Paper Award, IEEE Transactions on Software Engineering
OpenAI Researcher Access Program Grant (\$5,000)
Google Cloud Research Credits Grant (\$2,500)
Research Staff Excellence Awards (awarded to only 4 researchers university-wide), Singapore Management University
SMU SCIS PhD Research Excellence Award, Singapore Management University
SMU SCIS Dean’s List Award, Singapore Management University
- 2023 **Presidential Doctoral Fellowship**, Singapore Management University
SCIS Dean’s List Award, Singapore Management University
- 2022 **Honorable Mention Award**, ACSAC 2022
ACM SIGSOFT Distinguished Paper Award Nomination, ASE 2022
- 2020 **Outstanding Undergraduate Thesis** (Top 0.2% in province), Jiangsu Province, China
Outstanding Graduate (Top 5%), Yangzhou University
- 2019 & Before **National Encouragement Scholarship** (Top 3%), Ministry of Education of China
2nd Class Presidential Scholarship, Yangzhou University
1st Class Presidential Scholarship, Yangzhou University

ACADEMIC SERVICES

[Program Committee \(PC\)/Reviewer](#)

- 2026 **International Conference on the Foundations of Software Engineering** (FSE), PC in Ideas, Visions and Reflections Track
- 2026 **Workshop on Human-Centered AI for Software Engineering** (HumanAISE), PC
- 2026 **International Conference on AI-powered Software** (Alware), PC in Main Track
- 2026 **International Conference on Software Maintenance and Evolution** (ICSME), PC in Research Papers & Visions and Emerging Results Track
- 2025–2026 **International Conference on Software Engineering** (ICSE), Shadow PC
- 2024–2026 **Mining Software Repositories Conference** (MSR), Junior PC

- 2025–2026 **International Workshop on Green and Sustainable Software** (GREENS@ICSE), PC
- 2026 **European Conference on Computer Systems** (EuroSys), Shadow PC & Artifact Evaluation
- 2026 **International Conference on Evaluation and Assessment in Software Engineering** (EASE), PC in AI Models/Data Track
- 2025 **Annual Meeting of the Association for Computational Linguistics** (ACL), Reviewer
- 2025 **IEEE International Computer Software and Applications Conference** (COMPSAC), PC

Organizing Committee

- 2024 **International Workshop on Software Engineering and AI for Data Quality in Cyber-Physical Systems** (SEA4DQ@FSE), Web Chair

Journal Reviewer

- 2025–Present **IEEE Transactions on Dependable and Secure Computing** (TDSC)
- 2025–Present **Empirical Software Engineering** (EMSE)
- 2023–Present **ACM Transactions on Software Engineering and Methodology** (TOSEM)
- 2023–Present **IEEE Transactions on Software Engineering** (TSE)
- 2021–Present **Neurocomputing** (Software Section)

External Reviewer

- 2021–2025 **International Conference on Software Engineering** (ICSE)
- 2022–2026 **IEEE/ACM International Conference on Automated Software Engineering** (ASE)
- 2023–2025 **ACM International Conference on the Foundations of Software Engineering** (ESEC/FSE)
- 2023 **International Conference on Software Maintenance and Evolution** (ICSME)
- 2023 **International Workshop on LLMs for Code** (LLM4Code@ICSE)
- 2022–2023 **International Joint Conference on Artificial Intelligence** (IJCAI)
- 2022 **IEEE International Conference on Software Analysis, Evolution and Reengineering** (SANER)
- 2022 **International Conference on AI Engineering – Software Engineering for AI** (CAIN)
- 2022 **International Workshop on Software Engineering for Responsible AI** (SE4RAI@ICSE)
- 2022 **International Conference on Engineering of Complex Computer Systems** (ICECCS)

INVITED TALKS

- Dec 2025 **Shanghai Jiao Tong University**, Research Seminar
Optimizing and Fortifying AI Software through the Lens of Artifact Synthesis
- Nov 2025 **A-Mobile Workshop of ASE 2025**, Keynote, Invited by Guozhu Meng
Optimizing and Fortifying AI Software through the Lens of Artifact Synthesis
- Nov 2025 **Korea University**, Workshop on the Next Generation of CodeLLMs and Their Applications, Invited by Dongsun Kim
Optimizing and Fortifying AI Software via Program Synthesis
- Sep 2025 **Singapore Management University**, RISE-WASP Workshop, Invited by Lingxiao Jiang
Fortifying Autonomous Systems via Program Synthesis
- Nov 2023 **Singapore Management University**, Research Cluster Seminar, Invited by David Lo
Towards Compact and Efficient Language Models of Code via Compression
- Oct 2023 **North Carolina State University**, Guest Lecture, Invited by Bowen Xu
Enhancing Efficiency of Large Language Models of Code via Model Compression [\[Link\]](#)

TEACHING EXPERIENCE

Jan 2026 **Singapore Management University**, Teaching Assistant in CS706, Invited by David Lo
Software Mining and Analysis, Graduate Core Course

Oct 2023 **North Carolina State University**, Guest Lecture in CSC791, Invited by Bowen Xu
Enhancing Efficiency of Large Language Models of Code via Model Compression [[Link](#)]

Jan 2016 - Jun 2016 **Yangzhou University**, Teaching Assistant
Digital Signal Processing, Undergraduate Core Course

GRANT PROPOSAL WRITING EXPERIENCE

2025 - 2027 **ESG-based Responsible AI: Toward Green, Secure, and Compliant LLM Utilisation for Digital Service Development Process**
CSIRO - A*STAR Partnership Program, Singapore & Australia
Total Award: Over SGD \$500K

2024 - 2028 **TrustedSEERs: Trusted Intelligent Work Bots for Engineering Better Software**
National Research Foundation (NRF), Singapore
Total Award: Over SGD \$3.2M

2023 - 2025 **TitanCA: Vulnerability Detection using Large Code Models**
Government Technology Agency (GovTech), Singapore
Total Award: Over SGD \$3.8M

REFEREES

David Lo, ACM/IEEE/ASE Fellow, ACM Distinguished Speaker, OUB Chair Professor
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Zhou Yang, Canada CIFAR AI Chair Nominee, Amii Fellow, Assistant Professor
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